# nipah — Extracted Transmission Models

## Overview

A total of 41 transmission models were extracted from 16 articles that were flagged for containing extractable transmission models. The predominant model architecture was Compartmental (75.6%), with Airborne or close contact (34.1%) representing the most commonly modelled transmission route.

## Summary

| Metric | Value |
|---|---|
| Models extracted | 41 |
| Articles considered | 16 |
| Deterministic models | 18 (43.9%) |
| Stochastic models | 23 (56.1%) |
| Models with available code | 18 (43.9%) |

## Figure 1: Model Overview

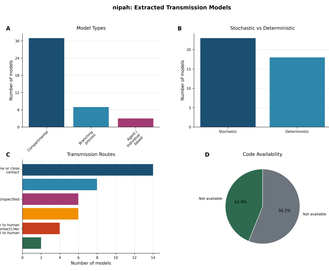


**Figure 1.** Overview of 41 transmission models extracted for nipah. (A) Distribution of model architectures. (B) Classification by stochastic or deterministic formulation. (C) Primary transmission routes incorporated. (D) Proportion of models with publicly available source code.

## Model Formulation

Deterministic formulations were employed in 43.9% of models, while stochastic approaches accounted for 56.1%. Empirical data were incorporated in 48.8% of the extracted models, with 36.6% classified as purely theoretical.

## Figure 2: Interventions and Assumptions

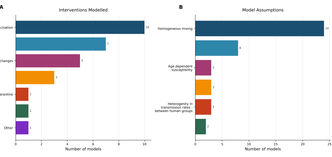


**Figure 2.** Interventions and assumptions in extracted nipah transmission models (n=41). (A) Frequency of intervention types evaluated across models. (B) Common modelling assumptions. Categories are not mutually exclusive; individual models may incorporate multiple interventions or assumptions.

## Interventions and Assumptions

The most frequently modelled interventions were Vaccination (24.4%), Treatment (17.1%), Behaviour changes (12.2%). Common modelling assumptions included Homogeneous mixing (58.5%), Age dependent susceptibility (7.3%), Heterogenity in transmission rates - between groups (7.3%).

## Figure 3: Programming Languages

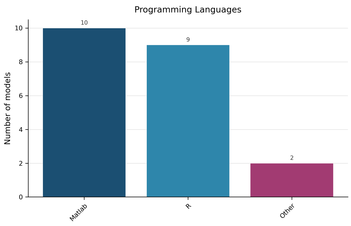


**Figure 3.** Programming languages used for model implementation among 21 models with specified language.

## Figure 4: Model Type by Formulation

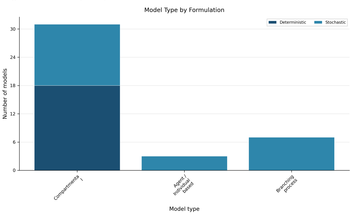


**Figure 4.** Distribution of deterministic and stochastic formulations across model types (n=41).

## Reproducibility

Source code was publicly available for 43.9% of the extracted models.

## Tables

### *Table 1: Model Types*

| Model Type | Count | Proportion |
|---|---|---|
| Compartmental | 31 | 75.6% |
| Branching process | 7 | 17.1% |
| Agent / Individual based | 3 | 7.3% |

*Distribution of model architectures across 41 extracted transmission models.*

### *Table 2: Model Formulation*

| Formulation | Count | Proportion |
|---|---|---|
| Stochastic | 23 | 56.1% |
| Deterministic | 18 | 43.9% |

*Classification of extracted models by stochastic versus deterministic formulation (n=41).*

### *Table 3: Transmission Routes*

| Transmission Route | Count | Proportion |
|---|---|---|
| Airborne or close contact | 14 | 34.1% |
| Vector/Animal to human;Human to human (direct contact) | 8 | 19.5% |
| Unspecified | 6 | 14.6% |
| Human to human (direct contact) | 6 | 14.6% |
| Human to human (direct contact);Vector/Animal to human | 4 | 9.8% |
| Vector/Animal to human;Human to human (direct non-sexual contact) | 2 | 4.9% |
| Vector/Animal to human | 1 | 2.4% |

*Primary transmission routes incorporated in extracted models (n=41).*

### *Table 4: Interventions Modelled*

| Intervention Type | Count | Proportion |
|---|---|---|
| Vaccination | 10 | 24.4% |
| Treatment | 7 | 17.1% |
| Behaviour changes | 5 | 12.2% |
| Vector/Animal control | 3 | 7.3% |
| Quarantine | 1 | 2.4% |
| Contact tracing | 1 | 2.4% |
| Other | 1 | 2.4% |

*Types of interventions evaluated in extracted models. Categories are not mutually exclusive (n=41).*

### *Table 5: Model Assumptions*

| Assumption | Count | Proportion |
|---|---|---|
| Homogeneous mixing | 24 | 58.5% |
| Other | 8 | 19.5% |
| Age dependent susceptibility | 3 | 7.3% |
| Heterogenity in transmission rates - between groups | 3 | 7.3% |
| Heterogenity in transmission rates - between human groups | 3 | 7.3% |
| Latent period is same as incubation period | 2 | 4.9% |

*Common modelling assumptions. Categories are not mutually exclusive (n=41).*

### *Table 6: Code Availability*

| Code Available | Count | Proportion |
|---|---|---|
| Yes | 18 | 43.9% |
| No | 23 | 56.1% |

*Availability of publicly accessible source code for extracted models (n=41).*

### *Table 7: Programming Languages*

| Programming Language | Count | Proportion |
|---|---|---|
| Unspecified | 20 | 48.8% |
| Matlab | 10 | 24.4% |
| R | 9 | 22.0% |
| Other | 2 | 4.9% |

*Programming languages used for model implementation (n=41).*

### *Table 8: Empirical Data Usage*

| Empirical Data Used | Count | Proportion |
|---|---|---|
| Yes | 20 | 48.8% |
| No/Unspecified | 21 | 51.2% |

## Model Inventory (Sample)

Table 9 presents a sample of 10 extracted model records. The complete dataset of 41 extracted transmission models for nipah is publicly available.

### *Table 9: Sample of Extracted Models*

| Article ID | Model Type | Compartmental Structure | Formulation | Transmission Route |
|---|---|---|---|---|
| PMID_33139552 | Compartmental | SIR | Deterministic | Unspecified |
| PMID_33139552 | Compartmental | SIR | Deterministic | Unspecified |
| PMID_33139552 | Compartmental | SIR | Deterministic | Unspecified |
| PMID_21632614 | Compartmental | Other compartmental | Deterministic | Unspecified |
| PMID_21632614 | Agent / Individual based | Not compartmental | Stochastic | Unspecified |
| DOI_325d66e3cbe7 | Compartmental | Other compartmental | Deterministic | Vector/Animal to human;Human to human |
| DOI_325d66e3cbe7 | Compartmental | Other compartmental | Deterministic | Vector/Animal to human;Human to human |
| PMID_11732410 | Compartmental | SEIR | Stochastic | Unspecified |
| DOI_1211690#edd2 | Compartmental | Other compartmental | Deterministic | Human to human (direct contact);Vector/A |
| PMID_35705986 | Branching process | Not compartmental | Stochastic | Vector/Animal to human;Human to human |

*Sample of 10 records from 41 extracted nipah transmission models.*

## Data Availability

The complete dataset of extracted transmission models for nipah (n=41) has been made publicly available to support future modelling efforts and outbreak preparedness.